Numerical Solution of the Sedov Self-Similar Blast Wave

Project I

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Abstract

The Sedov self-similar solution provides a fundamental description of the adiabatic expansion of a strong blast wave in a uniform medium. In this project, the dimensionless Euler equations governing the Sedov blast-wave solution were transformed into a system of coupled ordinary differential equations and solved numerically. Following the instructor's recommendation, the integration was performed using the sixth-order Runge–Kutta method of Luther (1968). The resulting dimensionless density, pressure, and velocity profiles were successfully reproduced and compared with the published Sedov solution. The similarity parameter was found to be $B = 1.1517$. In addition, the fractional contributions of kinetic and thermal energy to the total explosion energy were calculated as $T/E = 0.2828$ and $U_g/E = 0.7172$, respectively. The results demonstrate the characteristic structure of a strong adiabatic blast wave and confirm the consistency of the numerical implementation.

Contents

1 Introduction

Strong shock waves are among the most important phenomena in astrophysics and play a central role in supernova explosions, stellar winds, and other high-energy events. As a shock wave propagates through the surrounding medium, it compresses and heats the gas, producing a rapidly expanding blast wave.

One of the most important theoretical descriptions of such an explosion is the Sedov–Taylor solution [1, 2]. This solution describes the adiabatic expansion of a strong blast wave in a uniform medium under the assumption that the total explosion energy is conserved. Under these conditions, the hydrodynamic variables become self-similar and depend only on a dimensionless coordinate.

The self-similar formulation reduces the Euler equations to a system of ordinary differential equations describing the density, pressure, and velocity profiles within the remnant. Because of its physical significance and mathematical simplicity, the Sedov solution remains a standard problem in computational astrophysics.

The objective of this project is to reproduce the Sedov self-similar blast-wave solution through numerical integration of the governing dimensionless equations. Following the instructor's recommendation, the coupled differential equations were solved using the sixth-order Runge–Kutta method of Luther (1968). The resulting profiles were used to determine the similarity parameter B and the fractional contributions of kinetic and thermal energy to the total explosion energy.

2 Theoretical Background

2.1 Sedov–Taylor Blast Wave

The Sedov–Taylor solution describes the adiabatic expansion of a strong spherical blast wave into a uniform medium. It applies when the explosion energy is conserved and radiative losses are negligible. Under these assumptions, the shock radius depends only on the explosion energy E, the ambient density ρ_0, and time t. Dimensional analysis leads to the relation

$$R(t) = B\left(\frac{Et^2}{\rho_0}\right)^{1/5}, \tag{1}$$

where R is the shock radius and B is a dimensionless similarity parameter determined by the internal structure of the solution.

The Sedov solution is self-similar, meaning that the density, pressure, and velocity profiles retain the same shape throughout the expansion. As a result, the hydrodynamic variables can be expressed in terms of a single dimensionless coordinate rather than independent spatial and temporal variables.

2.2 Dimensionless Variables

The self-similar coordinate is defined as

$$\xi = \frac{r}{R}, \tag{2}$$

where r denotes the radial distance from the explosion centre and R is the instantaneous shock radius.

Following the Sedov formulation, the hydrodynamic variables are written in dimensionless form as

$$\rho = \frac{\gamma+1}{\gamma-1}\rho_0 F(\xi), \tag{3}$$

$$P_g = \frac{2}{\gamma+1}\rho_0 \dot{R}^2 G(\xi), \tag{4}$$

$$v = \frac{2}{\gamma+1}\dot{R}U(\xi). \tag{5}$$

Here $F(\xi)$, $G(\xi)$, and $U(\xi)$ represent the dimensionless density, pressure, and velocity profiles, respectively.

2.3 Dimensionless Euler Equations

The evolution of the blast wave is governed by the Euler equations expressing conservation of mass, momentum, and energy. After introducing the self-similar variables, the original partial differential equations are transformed into a system of coupled ordinary differential equations.

For an ideal gas with adiabatic index γ, the pressure function satisfies the adiabatic relation

$$G^{\gamma-1}Z = \text{constant}. \tag{6}$$

This relation allows the pressure variable to be eliminated from the system, reducing the problem to two coupled differential equations for the dimensionless density and velocity functions. These equations are solved numerically to obtain the internal structure of the Sedov blast wave.

3 Derivation of the Governing Equations

3.1 Dimensionless Euler Equations

Using the self-similar variables introduced in the previous section, the Euler equations reduce to the following dimensionless form [4]:

$$\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right)\xi\frac{F'}{F} + U' + \frac{2U}{\xi} = 0, \tag{7}$$

$$\xi\left[\left(\frac{2}{\gamma+1}\frac{U}{\xi} - 1\right)U' - \frac{3}{2}\frac{U}{\xi}\right] + \frac{\gamma-1}{\gamma+1}\frac{1}{F}G' = 0, \tag{8}$$

$$GF^{1-\gamma}\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right) = \frac{1-\gamma}{2\xi^3}. \tag{9}$$

The above equations represent the conservation of mass, momentum, and energy in self-similar form.

3.2 Final Form of the Differential Equations

By eliminating the pressure derivative using the adiabatic relation and combining the continuity and momentum equations, the system can be reduced to two coupled first-order ordinary differential equations for $U(\xi)$ and $F(\xi)$:

$$\frac{dU}{d\xi} = \frac{1}{\xi}\left[\frac{(\gamma-1)^2F^{\gamma-2}\left(\frac{3}{2}(\gamma+1)\xi - 2\gamma U\right)}{(\gamma-1)^2\gamma F^{\gamma-2} + 4\xi^2\left(U - \frac{\gamma+1}{2}\xi\right)^3} + \frac{3(\gamma+1)\xi^3U\left(U - \frac{\gamma+1}{2}\xi\right)^2}{(\gamma-1)^2\gamma F^{\gamma-2} + 4\xi^2\left(U - \frac{\gamma+1}{2}\xi\right)^3}\right], \tag{10}$$

$$\frac{dF}{d\xi} = -\frac{F}{U - \frac{\gamma+1}{2}\xi}\left[\frac{\frac{3}{2}(\gamma+1)(\gamma-1)^2F^{\gamma-2}}{(\gamma-1)^2\gamma F^{\gamma-2} + 4\xi^2\left(U - \frac{\gamma+1}{2}\xi\right)^3} + \frac{\xi U\left(8U - (\gamma+1)\xi\right)\left(U - \frac{\gamma+1}{2}\xi\right)^2}{(\gamma-1)^2\gamma F^{\gamma-2} + 4\xi^2\left(U - \frac{\gamma+1}{2}\xi\right)^3}\right]. \tag{11}$$

4 Numerical Method

4.1 Boundary Conditions

The self-similar solution is defined in the interval

$$0 \le \xi \le 1, \tag{12}$$

where $\xi = 1$ corresponds to the shock front. Following the project specification, the dimensionless variables satisfy the shock boundary conditions

$$F(1) = 1, \tag{13}$$

$$G(1) = 1, \tag{14}$$

$$U(1) = 1. \tag{15}$$

These conditions represent the normalized post-shock values of the density, pressure, and velocity. They provide the starting point for the numerical integration.

4.2 Classical Fourth-Order Runge–Kutta Method

The original project description proposed the use of the classical fourth-order Runge–Kutta (RK4) method for solving the coupled ordinary differential equations. For a system written as

$$y' = f(x, y, z), \tag{16}$$

$$z' = g(x, y, z), \tag{17}$$

the RK4 method evaluates the derivatives at several intermediate points within each integration step and combines them to obtain a fourth-order accurate approximation to the solution.

The method provides a robust and widely used approach for solving coupled ordinary differential equations arising in self-similar hydrodynamic problems.

4.3 Luther Sixth-Order Runge–Kutta Scheme

Following the instructor's recommendation, the numerical calculations were ultimately performed using the sixth-order Runge–Kutta method proposed by Luther (1968) [3]. This method extends the classical Runge–Kutta approach by employing additional intermediate stages, thereby improving the accuracy of the numerical integration.

Compared with the standard RK4 scheme, the Luther method achieves higher-order accuracy while preserving the explicit structure of the integration algorithm. The method is particularly suitable for problems involving rapidly varying solutions near the shock front, where increased numerical precision is desirable.

The numerical implementation used seven intermediate stages and advanced the solution according to the Luther sixth-order coefficients.

4.4 Numerical Integration Procedure

The integration was performed from the shock front ($\xi = 1$) toward the centre of the remnant. The initial conditions

$$F(1) = 1, \tag{18}$$

$$U(1) = 1, \tag{19}$$

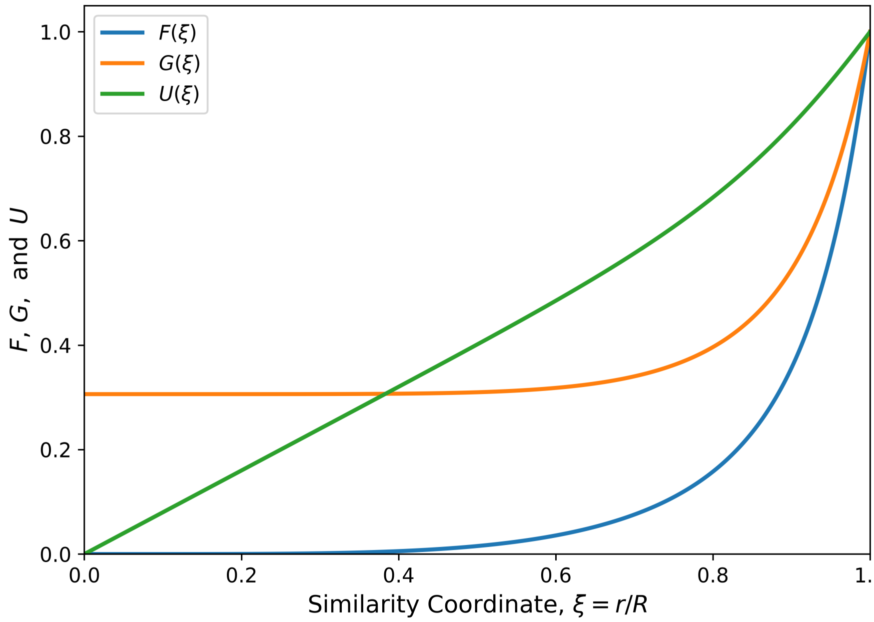

Figure 1: Dimensionless density $F(\xi)$, pressure $G(\xi)$, and velocity $U(\xi)$ profiles obtained from the numerical solution of the Sedov self-similar blast-wave equations for $\gamma = 5/3$.

were used to start the calculation.

At each integration step, the dimensionless density and velocity profiles were updated using the Luther sixth-order Runge–Kutta scheme. Once the functions $F(\xi)$ and $U(\xi)$ had been obtained, the dimensionless pressure profile $G(\xi)$ was reconstructed from the adiabatic relation.

The resulting profiles were subsequently used to evaluate the similarity parameter B and the fractional contributions of kinetic and thermal energy through numerical integration of the corresponding energy expressions.

5 Results and Discussion

5.1 Dimensionless Shock Profiles

The coupled differential equations were solved numerically using the sixth-order Runge–Kutta method. The resulting dimensionless density, pressure, and velocity profiles are shown in Figure 1.

The numerical solution successfully reproduces the characteristic structure of the Sedov blast wave. The density profile approaches zero near the centre of the remnant and increases rapidly toward the shock front. This behaviour reflects the strong compression of the gas immediately behind the shock.

The pressure profile remains finite throughout the remnant interior and increases toward the shock boundary. The smooth pressure distribution is consistent with the adiabatic nature of the Sedov solution, where the explosion energy is conserved and redistributed throughout the expanding gas.

The velocity profile increases monotonically with the similarity coordinate and reaches its normalized post-shock value at $\xi = 1$. Near the centre of the remnant, the velocity approaches zero, indicating that the gas motion becomes progressively weaker toward the origin.

Overall, the obtained profiles are in good agreement with the published Sedov solution and confirm the correctness of the numerical implementation.

5.2 Similarity Parameter

The similarity parameter B determines the normalization of the shock radius through the relation

$$R(t) = B \left(\frac{Et^2}{\rho_0} \right)^{1/5}. \tag{20}$$

Using the numerical solution and the energy integral, the parameter was found to be

$$B = 1.1517. \tag{21}$$

The project statement quotes a value of approximately $B \approx 1.15$ for the Sedov solution. The excellent agreement between the numerical and theoretical values demonstrates the accuracy of the integration procedure and confirms that the self-similar solution has been reproduced successfully.

5.3 Energy Distribution

The kinetic and thermal energy fractions were calculated using the numerical solution and the corresponding energy integrals. The results are summarized in Table 1.

Table 1: Numerical results obtained from the Sedov solution.

Quantity	Value
Similarity parameter B	1.1517
Kinetic energy fraction T/E	0.2828
Thermal energy fraction U_g/E	0.7172

The results indicate that approximately 28.3% of the explosion energy remains in kinetic form, while 71.7% is stored as thermal energy of the shocked gas. Therefore, the thermal component represents the dominant energy reservoir during the Sedov phase.

This behaviour is expected for a strong adiabatic shock. As the blast wave propagates through the surrounding medium, a significant fraction of the explosion energy is converted into internal energy through shock heating. Consequently, the thermal energy exceeds the kinetic energy throughout the evolution described by the Sedov solution.

The obtained energy fractions are physically reasonable and consistent with the expected properties of an energy-conserving blast wave.

6 Conclusion

In this project, the Sedov self-similar blast-wave solution was reproduced through the numerical integration of the dimensionless Euler equations. The governing equations were transformed into a coupled system of ordinary differential equations and solved using the sixth-order Runge–Kutta method of Luther (1968).

The numerical solution successfully reproduced the characteristic density, pressure, and velocity profiles of the Sedov blast wave. The obtained profiles showed good agreement with the expected behaviour of a strong adiabatic shock propagating through a uniform medium.

Using the numerical solution, the similarity parameter was determined to be

$$B = 1.1517, \tag{22}$$

which is in excellent agreement with the theoretical value quoted for the Sedov solution. The energy analysis further showed that approximately 28.3% of the explosion energy is stored in kinetic form, while 71.7% is contained in the thermal energy of the shocked gas.

Overall, the results confirm the validity of the numerical implementation and demonstrate the effectiveness of self-similar methods for studying astrophysical blast waves.

A Numerical Implementation

A.1 Physical Parameters

```python
import numpy as np


gamma = 5.0/3.0

A = (gamma + 1)/2
```

Listing 1: Physical parameters used in the Sedov blast-wave calculations.

A.2 Implementation of the Governing Equations

```python
def denominator(xi, F, U):

    return (
        (gamma - 1)**2 * gamma * F**(gamma - 2)
        +
        4*xi**2*(U - A*xi)**3
    )


def dU_dxi(xi, F, U):

    D = denominator(xi, F, U)

    term1 = (
        (gamma - 1)**2
        * F**(gamma - 2)
        * (
            1.5*(gamma + 1)*xi
            - 2*gamma*U
        )
    )

    term2 = (
        3*(gamma + 1)
        * xi**3
        * U
        * (U - A*xi)**2
    )

    return (1/xi)*(term1/D + term2/D)


def dF_dxi(xi, F, U):
```

```
    D = denominator(xi, F, U)

    part1 = (
        1.5*(gamma + 1)
        * (gamma - 1)**2
        * F**(gamma - 2)
    )

    part2 = (
        xi * U
        * (8*U - (gamma + 1)*xi)
        * (U - A*xi)**2
    )

    bracket = part1/D + part2/D

    return -F/(U - A*xi)*bracket


def system(xi, y):

    F = y[0]
    U = y[1]

    return np.array([
        dF_dxi(xi, F, U),
        dU_dxi(xi, F, U)
    ])
```

Listing 2: Functions defining the dimensionless Sedov equations.

A.3 Luther Sixth-Order Runge–Kutta Integrator

```
sqrt21 = np.sqrt(21.0)

def rk6_step(x, y, h):

    k1 = h*system(x, y)

    k2 = h*system(
        x + h,
        y + k1
    )

    k3 = h*system(
        x + h/2,
        y + (3*k1 + k2)/8
```

```
    )

    k4 = h*system(
        x + 2*h/3,
        y + (8*k1 + 2*k2 + 8*k3)/27
    )

    k5 = h*system(
        x + (7 - sqrt21)*h/14,
        y + (
            3*(3*sqrt21 - 7)*k1
            - 8*(7 - sqrt21)*k2
            + 48*(7 - sqrt21)*k3
            - 3*(21 - sqrt21)*k4
        )/392
    )

    k6 = h*system(
        x + (7 + sqrt21)*h/14,
        y + (
            -5*(231 + 51*sqrt21)*k1
            -40*(7 + sqrt21)*k2
            -320*sqrt21*k3
            +3*(21 + 121*sqrt21)*k4
            +392*(6 + sqrt21)*k5
        )/1960
    )

    k7 = h*system(
        x + h,
        y + (
            15*(22 + 7*sqrt21)*k1
            +120*k2
            +40*(7*sqrt21 - 5)*k3
            -63*(3*sqrt21 - 2)*k4
            -14*(49 + 9*sqrt21)*k5
            +70*(7 - sqrt21)*k6
        )/180
    )

    y_next = y + (
        9*k1
        +64*k3
        +49*k5
        +49*k6
        +9*k7
    )/180

```

```
    return y_next
```

Listing 3: Implementation of the Luther (1968) sixth-order Runge–Kutta method.

A.4 Numerical Integration

```
h = -1e-5

xi_values = [1.0]
F_values = [1.0]
U_values = [1.0]

y = np.array([1.0, 1.0])

xi = 1.0

while xi > 1e-4:

    y = rk6_step(xi, y, h)

    xi += h

    xi_values.append(xi)

    F_values.append(y[0])

    U_values.append(y[1])
```

Listing 4: Numerical integration of the Sedov equations.

A.5 Pressure Reconstruction and Energy Calculations

```
G_values = []

for xi, F, U in zip(xi_values, F_values, U_values):

    rhs = (1 - gamma)/(2*xi**3)

    factor = (
        F**(1-gamma)
        *
        (U/xi - (gamma+1)/2)
    )

    G = rhs/factor

    G_values.append(G)
```

```

xi_values = np.array(xi_values)
F_values = np.array(F_values)
G_values = np.array(G_values)
U_values = np.array(U_values)

kinetic_integrand = (
    (1/(gamma-1))
    * F_values
    * U_values**2
    * xi_values**2
)

thermal_integrand = (
    (1/(gamma-1))
    * G_values
    * xi_values**2
)

T = abs(np.trapezoid(
    kinetic_integrand,
    xi_values
))

Ug = abs(np.trapezoid(
    thermal_integrand,
    xi_values
))

I = T + Ug

T_frac = T/total

Ug_frac = Ug/total

integrand = (
    (1/(gamma-1))
    * F_values
    * U_values**2
    +
    (1/(gamma-1))
    * G_values
)

integrand *= xi_values**2

I = abs(
    np.trapezoid(
```

```
        integrand,
        xi_values
    )
)

B_minus5 = (
    32*np.pi
    /
    (25*(gamma+1))
) * I

B = (1/B_minus5)**(1/5)
```

Listing 5: Reconstruction of pressure and evaluation of the energy integrals.

References

[1] L. I. Sedov, *Similarity and Dimensional Methods in Mechanics*, Academic Press, New York, 1959.

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